GED Math Practice Test 2

PART A — QUESTIONS

GED-MATH2-001
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A worker fills 3/5 of a bin with packing material. Which decimal is equal to 3/5?
A) 0.35
B) 0.53
C) 0.6
D) 1.67

GED-MATH2-002
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Non-Calculator
Prompt:
What is the value of $4a + 2$ when $a = 6$?
A) 18
B) 22
C) 26
D) 32

GED-MATH2-003
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A class has 20 students. If 30% of the students ride the bus, how many students ride the bus?
A) 3
B) 6
C) 10
D) 14

GED-MATH2-004
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Non-Calculator
Prompt:
Which expression is equivalent to $7x - 3x + 5$?
A) $4x + 5$
B) $10x + 5$
C) $4x - 5$
D) $7x + 2$

GED-MATH2-005
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A fruit punch recipe uses 2 cups of juice for every 3 cups of sparkling water. If 6 cups of sparkling water are used, how many cups of juice are needed?
A) 2
B) 4
C) 6
D) 9

GED-MATH2-006
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A moving company charges $85 plus $0.65 per mile. Which expression gives the total charge, in dollars, for a move of m miles?
A) 85m + 0.65
B) 0.65m + 85
C) 85.65m
D) 0.65(m + 85)

GED-MATH2-007
Section: Math
Type: numeric_entry
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A hallway is 14 yards long. How many feet long is the hallway?
Type your answer as an integer.

GED-MATH2-008
Section: Math
Type: dropdown
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula W = 40 + 8d gives a worker's total weekly pay W, in dollars, for completing d deliveries plus a weekly bonus. In this formula, 40 represents [D1], and 8 represents [D2].
Dropdowns:
D1) the pay per delivery | the weekly bonus | the number of deliveries
D2) the pay per delivery | the weekly bonus | the total weekly pay

GED-MATH2-009
Section: Math
Type: mcq
Category: Quantitative Problem Solving

Calculator: Calculator-Allowed
Prompt:
A train travels 315 miles in 5 hours. What is the average speed of the train?
A) 53 miles per hour
B) 60 miles per hour
C) 63 miles per hour
D) 68 miles per hour

GED-MATH2-010
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for k: $6k - 14 = 34$.
A) 6
B) 8
C) 10
D) 12

GED-MATH2-011
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows the high temperature, in degrees Fahrenheit, for 5 days.

Day: Monday, Tuesday, Wednesday, Thursday, Friday
Temperature: 71, 76, 69, 82, 78

What is the range of the temperatures?
A) 9°F
B) 11°F
C) 13°F
D) 15°F

GED-MATH2-012
Section: Math
Type: multi_select
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
Which inequalities are true when $x = 4$?
A) $3x + 2 > 13$
B) $5x - 1 \leq 18$
C) $2x + 7 = 15$
D) $x/2 + 6 \geq 8$
E) $4x - 9 < 5$

GED-MATH2-013
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A snack box contains 4 granola bars, 6 fruit cups, and 10 crackers. If one item is chosen at random, what is the probability that it is a fruit cup?
A) 1/5
B) 3/10
C) 2/5
D) 3/5

GED-MATH2-014
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for x: $7x + 18 = 74$.
Type your answer as an integer.

GED-MATH2-015
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A rectangular patio is 15 feet long and 9 feet wide. What is the perimeter of the patio?
A) 24 feet
B) 48 feet
C) 72 feet
D) 135 feet

GED-MATH2-016
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows a linear relationship.

x: 2, 4, 6, 8
y: 9, 15, 21, 27

Which equation represents the relationship?
A) $y = 2x + 5$
B) $y = 3x + 3$
C) $y = 4x + 1$
D) $y = 6x - 3$

GED-MATH2-017
Section: Math
Type: multi_select
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
Which statements are true?
A) 2.5 meters is equal to 250 centimeters.
B) 4 liters is equal to 400 milliliters.
C) 3 kilograms is equal to 3,000 grams.
D) 0.75 kilometer is equal to 75 meters.
E) 1,200 grams is equal to 12 kilograms.

GED-MATH2-018
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve the literal equation $V = lwh$ for h.
A) $h = Vlw$
B) $h = V/(lw)$
C) $h = lw/V$
D) $h = V/l + w$

GED-MATH2-019
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A store increases the price of a lamp from $40 to $46. What is the percent increase?
A) 6%
B) 10%
C) 12%
D) 15%

GED-MATH2-020
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A function is defined by $f(x) = 5x - 11$. What is $f(9)$?
Type your answer as an integer.

GED-MATH2-021
Section: Math
Type: mcq
Category: Quantitative Problem Solving

Calculator: Calculator-Allowed
Prompt:
A ramp rises 8 inches for every 15 inches of horizontal distance. What is the length of the ramp?
A) 17 inches
B) 19 inches
C) 23 inches
D) 25 inches

GED-MATH2-022
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A savings account balance grows by 4% each year. If the starting balance is \$1,200, which expression gives the balance after 3 years?
A) $1{,}200(0.04)^3$
B) $1{,}200(0.96)^3$
C) $1{,}200(1.04)^3$
D) $1{,}200 + 1.04^3$

GED-MATH2-023
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A circular table has a radius of 3 feet. Using $\pi \approx 3.14$, what is the area of the tabletop?
A) 9.42 square feet
B) 18.84 square feet
C) 28.26 square feet
D) 56.52 square feet

GED-MATH2-024
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A theater sold adult tickets for \$12 each and student tickets for \$8 each. A total of 40 tickets were sold for \$400. Which system can be used to find a, the number of adult tickets, and s, the number of student tickets?
A) $a + s = 400$
$12a + 8s = 40$
B) $a + s = 40$
$12a + 8s = 400$
C) $12 + 8 = 40$
$a + s = 400$
D) $8a + 12s = 40$
$a + s = 400$

GED-MATH2-025
Section: Math
Type: numeric_entry
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
The weights, in pounds, of five packages are 8, 12, 15, 15, and 25. What is the mean weight? Type your answer as a decimal rounded to the nearest tenth.

GED-MATH2-026
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Company A charges $25 plus $4 per hour for equipment rental. Company B charges $10 plus $7 per hour. For how many hours will the two companies charge the same amount?
A) 3 hours
B) 4 hours
C) 5 hours
D) 6 hours

GED-MATH2-027
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
On a scale drawing, 1 centimeter represents 8 feet. A storage room is 6.25 centimeters long on the drawing. What is the actual length of the storage room?
A) 14.25 feet
B) 48 feet
C) 50 feet
D) 62.5 feet

GED-MATH2-028
Section: Math
Type: dropdown
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The equation $y = -3x + 18$ is written in slope-intercept form. The slope is [D1], and the y-intercept is [D2].
Dropdowns:
D1) −3 | 3 | 18
D2) −3 | 3 | 18

GED-MATH2-029
Section: Math
Type: mcq
Category: Quantitative Problem Solving

Calculator: Calculator-Allowed
Prompt:
A customer buys 3.2 pounds of coffee at $7.25 per pound. What is the total cost?
A) $10.45
B) $20.40
C) $23.20
D) $27.50

GED-MATH2-030
Section: Math
Type: multi_select
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
Which statements are true about the function $h(x) = x^2 - 4$?
A) $h(0) = -4$
B) $h(2) = 0$
C) $h(3) = 5$
D) $h(-2) = -8$
E) $h(4) = 12$

GED-MATH2-031
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A family's water bill is $64.80 for using 1,800 gallons of water. What is the cost per gallon?
A) $0.036
B) $0.36
C) $2.78
D) $27.78

GED-MATH2-032
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for r: $4(r + 6) = 52$.
A) 7
B) 10
C) 13
D) 19

GED-MATH2-033
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed

Prompt:
A fish tank is shaped like a rectangular prism. It is 30 inches long, 12 inches wide, and 16 inches high. What is the volume of the tank?
A) 720 cubic inches
B) 1,856 cubic inches
C) 5,760 cubic inches
D) 11,520 cubic inches

GED-MATH2-034
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A rental car costs $45 per day plus $0.20 per mile. If d is the number of days and m is the number of miles, which expression represents the total cost?
A) 45d + 0.20m
B) 45m + 0.20d
C) 45 + 0.20dm
D) 45(d + m)

GED-MATH2-035
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which verbal statement matches the expression 9 − 2x?
A) Nine less than twice a number
B) Twice the difference of nine and a number
C) Nine decreased by twice a number
D) Twice a number decreased by nine

GED-MATH2-036
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A box contains 24 cans. Each can holds 12 ounces of soup. How many total ounces of soup are in the box?
A) 144
B) 216
C) 288
D) 360

GED-MATH2-037
Section: Math
Type: drag_drop
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed

Prompt:
Match each target with the correct tile.
Tiles:
A) $2x + 6 = 20$
B) $20 - 2x = 6$
C) $6x + 2 = 20$
D) $2(x + 6) = 20$
Targets:
T1) Twice a number plus 6 is 20
T2) Six times a number plus 2 is 20
T3) Twice the sum of a number and 6 is 20

GED-MATH2-038
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A store sells a printer for $180. During a sale, the price is reduced by 18%. What is the sale price?
A) $32.40
B) $147.60
C) $162.00
D) $212.40

GED-MATH2-039
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which ordered pair is a solution to the equation $y = 2x - 5$?
A) (2, 1)
B) (3, 1)
C) (4, 5)
D) (5, 15)

GED-MATH2-040
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A quadratic function is defined by $q(x) = x^2 + 3x$. What is $q(5)$?
A) 10
B) 25
C) 40
D) 55

GED-MATH2-041
Section: Math
Type: multi_select

Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
A grocery receipt shows the following item costs:

$4.50, $6.25, $3.75, $8.00

Which statements are true?
A) The total cost is $22.50.
B) The most expensive item costs $8.00.
C) The range of the item costs is $5.25.
D) The mean item cost is $6.25.
E) The total cost is $21.50.

GED-MATH2-042
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A line passes through the points (2, 7) and (6, 19). What is the slope of the line?
A) 2
B) 3
C) 4
D) 12

GED-MATH2-043
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A printing service charges a setup fee plus a cost per flyer. The table shows the total cost.

Flyers: 50, 100, 150, 200
Total cost: $42, $69, $96, $123

What is the cost per additional flyer?
A) $0.27
B) $0.54
C) $1.23
D) $27.00

GED-MATH2-044
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which expression is equivalent to $5(2x - 3) + 4x$?

A) $6x - 15$
B) $10x - 11$
C) $14x - 15$
D) $14x + 15$

GED-MATH2-045
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve the system of equations.

$2x + y = 17$
$x + y = 11$

What is the value of y?
A) 3
B) 5
C) 6
D) 9

GED-MATH2-046
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A community garden has 480 square feet of planting space. If 35% of the space is used for vegetables, how many square feet are used for vegetables?
A) 128
B) 156
C) 168
D) 312

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH2-001 — Correct: C — Correct Answer: 0.6 — Explanation: Divide 3 by 5 to get 0.6.

GED-MATH2-002 — Correct: C — Correct Answer: 26 — Explanation: Substitute $a = 6$: $4(6) + 2 = 24 + 2 = 26$.

GED-MATH2-003 — Correct: B — Correct Answer: 6 — Explanation: 30% of 20 is $0.30 \times 20 = 6$.

GED-MATH2-004 — Correct: A — Correct Answer: $4x + 5$ — Explanation: Combine like terms: $7x - 3x + 5 = 4x + 5$.

GED-MATH2-005 — Correct: B — Correct Answer: 4 — Explanation: The ratio juice to sparkling water is 2:3. Doubling 3 cups of sparkling water to 6 cups requires doubling 2 cups of juice to 4 cups.

GED-MATH2-006 — Correct: B — Correct Answer: $0.65m + 85$ — Explanation: The total charge is the fixed \$85 fee plus \$0.65 for each mile.

GED-MATH2-007 — Correct: 42 — Tolerance: exact — Correct Answer: 42 — Explanation: There are 3 feet in 1 yard, so 14 yards = $14 \times 3 = 42$ feet.

GED-MATH2-008 — Correct: D1=the weekly bonus; D2=the pay per delivery — Correct Answer: D1=the weekly bonus; D2=the pay per delivery — Explanation: The constant 40 is the weekly bonus, and the coefficient 8 is the amount paid per delivery.

GED-MATH2-009 — Correct: C — Correct Answer: 63 miles per hour — Explanation: Average speed is distance divided by time: $315 \div 5 = 63$ miles per hour.

GED-MATH2-010 — Correct: B — Correct Answer: 8 — Explanation: Add 14 to both sides to get $6k = 48$, then divide by 6 to get $k = 8$.

GED-MATH2-011 — Correct: C — Correct Answer: 13°F — Explanation: The range is the greatest value minus the least value: $82 - 69 = 13$.

GED-MATH2-012 — Correct: A,D — Correct Answer: $3x + 2 > 13$ | $x/2 + 6 \geq 8$ — Explanation: When $x = 4$, A gives $14 > 13$ and D gives $8 \geq 8$.

GED-MATH2-013 — Correct: B — Correct Answer: 3/10 — Explanation: There are 20 total items, and 6 are fruit cups. The probability is $6/20 = 3/10$.

GED-MATH2-014 — Correct: 8 — Tolerance: exact — Correct Answer: 8 — Explanation: Subtract 18 from both sides to get $7x = 56$, then divide by 7 to get $x = 8$.

GED-MATH2-015 — Correct: B — Correct Answer: 48 feet — Explanation: Perimeter = 2(length + width) = $2(15 + 9) = 48$ feet.

GED-MATH2-016 — Correct: B — Correct Answer: $y = 3x + 3$ — Explanation: The y-values increase by 6 when x increases by 2, so the rate is 3. Substituting $x = 2$ and $y = 9$ gives $y = 3x + 3$.

GED-MATH2-017 — Correct: A,C — Correct Answer: 2.5 meters is equal to 250 centimeters. | 3 kilograms is equal to 3,000 grams. — Explanation: 1 meter = 100 centimeters, so 2.5 meters = 250 centimeters. Also, 1 kilogram = 1,000 grams, so 3 kilograms = 3,000 grams.

GED-MATH2-018 — Correct: B — Correct Answer: $h = V/(lw)$ — Explanation: Divide both sides of $V = lwh$ by lw to solve for h.

GED-MATH2-019 — Correct: D — Correct Answer: 15% — Explanation: The increase is \$46 − \$40 = \$6. Then $6 \div 40 = 0.15$, or 15%.

GED-MATH2-020 — Correct: 34 — Tolerance: exact — Correct Answer: 34 — Explanation: $f(9) = 5(9) - 11 = 45 - 11 = 34$.

GED-MATH2-021 — Correct: A — Correct Answer: 17 inches — Explanation: Use the Pythagorean theorem: $8^2 + 15^2 = 64 + 225 = 289$, and $\sqrt{289} = 17$.

GED-MATH2-022 — Correct: C — Correct Answer: $1{,}200(1.04)^3$ — Explanation: A 4% annual increase means the balance is multiplied by 1.04 each year.

GED-MATH2-023 — Correct: C — Correct Answer: 28.26 square feet — Explanation: Area = $\pi r^2 = 3.14(3^2) = 3.14(9) = 28.26$ square feet.

GED-MATH2-024 — Correct: B — Correct Answer: $a + s = 40$
$12a + 8s = 400$ — Explanation: The number of tickets totals 40, and the money from adult and student tickets totals $400.

GED-MATH2-025 — Correct: 15.0 — Tolerance: exact — Correct Answer: 15.0 — Explanation: The total weight is $8 + 12 + 15 + 15 + 25 = 75$. Divide by 5 to get 15.0 pounds.

GED-MATH2-026 — Correct: C — Correct Answer: 5 hours — Explanation: Set the costs equal: $25 + 4h = 10 + 7h$. Then $15 = 3h$, so $h = 5$.

GED-MATH2-027 — Correct: C — Correct Answer: 50 feet — Explanation: Multiply 6.25 centimeters by 8 feet per centimeter: $6.25 \times 8 = 50$ feet.

GED-MATH2-028 — Correct: D1=−3; D2=18 — Correct Answer: D1=−3; D2=18 — Explanation: In $y = mx + b$, m is the slope and b is the y-intercept.

GED-MATH2-029 — Correct: C — Correct Answer: $23.20 — Explanation: Multiply 3.2 by $7.25 to get $23.20.

GED-MATH2-030 — Correct: A,B — Correct Answer: $h(0) = -4$ | $h(2) = 0$ — Explanation: $h(0) = 0^2 - 4 = -4$, and $h(2) = 2^2 - 4 = 0$.

GED-MATH2-031 — Correct: A — Correct Answer: $0.036 — Explanation: Divide the bill by the gallons used: $64.80 \div 1{,}800 = 0.036$.

GED-MATH2-032 — Correct: A — Correct Answer: 7 — Explanation: Divide both sides by 4 to get $r + 6 = 13$, then subtract 6 to get $r = 7$.

GED-MATH2-033 — Correct: C — Correct Answer: 5,760 cubic inches — Explanation: Volume = length × width × height = $30 \times 12 \times 16 = 5{,}760$ cubic inches.

GED-MATH2-034 — Correct: A — Correct Answer: $45d + 0.20m$ — Explanation: The cost is $45 for each day plus $0.20 for each mile.

GED-MATH2-035 — Correct: C — Correct Answer: Nine decreased by twice a number — Explanation: The expression $9 - 2x$ means 9 minus twice a number.

GED-MATH2-036 — Correct: C — Correct Answer: 288 — Explanation: Multiply 24 cans by 12 ounces per can: $24 \times 12 = 288$ ounces.

GED-MATH2-037 — Correct: T1=A; T2=C; T3=D — Correct Answer: T1=$2x + 6 = 20$; T2=$6x + 2 = 20$; T3=$2(x + 6) = 20$ — Explanation: Each equation matches the order and grouping described in the verbal statement.

GED-MATH2-038 — Correct: B — Correct Answer: \$147.60 — Explanation: An 18% discount on \$180 is \$32.40, so the sale price is \$180 − \$32.40 = \$147.60.

GED-MATH2-039 — Correct: B — Correct Answer: (3, 1) — Explanation: Substitute $x = 3$: $y = 2(3) - 5 = 1$.

GED-MATH2-040 — Correct: C — Correct Answer: 40 — Explanation: $q(5) = 5^2 + 3(5) = 25 + 15 = 40$.

GED-MATH2-041 — Correct: A,B — Correct Answer: The total cost is \$22.50. | The most expensive item costs \$8.00. — Explanation: The total is 4.50 + 6.25 + 3.75 + 8.00 = 22.50, and the greatest cost is \$8.00.

GED-MATH2-042 — Correct: B — Correct Answer: 3 — Explanation: Slope = $(19 - 7)/(6 - 2) = 12/4 = 3$.

GED-MATH2-043 — Correct: B — Correct Answer: \$0.54 — Explanation: The cost increases by \$27 when the number of flyers increases by 50, so the cost per flyer is 27 ÷ 50 = \$0.54.

GED-MATH2-044 — Correct: C — Correct Answer: $14x - 15$ — Explanation: Distribute first: $5(2x - 3) + 4x = 10x - 15 + 4x = 14x - 15$.

GED-MATH2-045 — Correct: B — Correct Answer: 5 — Explanation: Subtract the second equation from the first to get $x = 6$. Then $6 + y = 11$, so $y = 5$.

GED-MATH2-046 — Correct: C — Correct Answer: 168 — Explanation: 35% of 480 is $0.35 \times 480 = 168$.